

MTH209: Practice Problems

Assignment A: Debugging and Indexing

1. Given an $n \times n$ matrix A , define a vector

$$r_k = \sum_i i = 1^k \sum_j j = 1^n a_{ij}.$$

Fix all errors in the following function so that it returns the correct vector.

```
rowTelescope <- function(A)
{
  n <- nrow(B)
  r <- numeric(n)
  for(k in 1:n)
  {
    r[k] <- sum(A[1:k])
  }
  return(r)
}
```

2. Write a function that takes a square matrix A and returns a vector t such that

$$t_k = \sum_{i=1}^k \sum_{j=1}^i a_{ij}.$$

```
lowerTriSum <- function(A)
{
  # your code here
}
```

3. For a square matrix A return a vector d such that

$$d_1 = a_{11}, \quad d_k = a_{kk} - a_{k-1,k-1}, \quad k \geq 2.$$

```
diagDiff <- function(A)
{
  # your code here
}
```

4. Redo Assignment 3

5. Write a function that takes a **diagonal matrix** D and returns D^{-1} .
6. For a given square invertible matrix A , write down its inverse using 2-3 methods.
7. Given a covariance matrix Σ and an integer k , return the $d \times k$ matrix of leading eigenvectors.
8. Given a vector x , return $\Sigma^{-1/2}x$.

```
whiten <- function(Sigma, x)
{
  # your code here
}
```

9. Transform a realization from $N(\mu, \Sigma)$ to a realization from $N(0, I)$

```
gauss_standardize <- function(x, mu, Sigma)
{
  # your code here
}
```

10. Write a function that calculates the Mahalanobis distance of a random vector x with mean μ and covariance Σ .

$$d(x) = (x - \mu)^\top \Sigma^{-1} (x - \mu)$$

Worksheet 5

1. Fit t-SNE with perplexity = 40 and dimension = 3 on the wine dataset. Return a vector of length two where:
 - a. The first element is the Euclidean distance between the 10th and 30th observations.
 - b. The second element is the **average Euclidean distance** between the 25th observation and all observations of `type = 3`
2. Using the **iris** dataset:
 - a. Run 2D t-SNE for perplexity values in $\{5, 10, 30, 50\}$.
 - b. Plot each embedding.
 - c. Quantify cluster separation by computing:
 - Average within-species pairwise distance
 - Average between-species pairwise distance
 - d. Report how this ratio changes with perplexity.
3. Using the wine dataset:
 - a. Implement PCA manually using eigen-decomposition.
 - b. Implement PCA using `prcomp()`.
 - c. Compare:
 - First two eigenvalues
 - First two principal component directions

- d. Compute the maximum absolute difference between the two PC1 score vectors.
4. Using the iris dataset:
 - a. Run 2D t-SNE with perplexity = 30.
 - b. Extract the embedding matrix.
 - c. Compute:
 - Distance between observation 10 and 50
 - Minimum distance from observation 10 to any observation of species “virginica”
 - d. Return a vector called **answer**.
5. Using the iris dataset:
 - a. Run t-SNE with perplexity values 5, 20, and 50.
 - b. For each embedding:
 - Compute average pairwise Euclidean distance.
 - Compute maximum pairwise Euclidean distance.
 - c. Store results in a matrix with rows corresponding to perplexity.
6. Using the wine dataset:
 - a. Compute the full pairwise distance matrix in the original feature space.
 - b. Compute pairwise distance matrices for:
 - 2D PCA
 - 2D t-SNE
 - 2D UMAP
 - c. For each method:
 - Compute correlation between original distances and embedding distances.
 - d. Report the three correlations.

Worksheet 6

1. For each distribution below:
 - Normal(0,1)
 - t3
 - t1
 - Gamma(shape = 2, rate = 1)
 - a. For n = 10, 50, 200, 1000:
 - Simulate one sample.
 - Compute cumulative sample means.
 - Plot the cumulative means against index.
 - b. Comment on stability as n increases.
2. Let $X_i \sim t_3$.
 - a. For n = 10, 50, 200, 1000:
 - Simulate 10,000 replicated sample means.
 - Estimate $P(|\bar{X}_n - 0| > 0.2)$.
 - b. Plot this probability against n.
 - c. Repeat for t1.

- d. Compare behavior.
3. Let $X_i \sim N(0, 1)$.
 - a. For $n = 5, 10, 30, 100$:
 - Simulate 10,000 samples.
 - Compute:
 - Unbiased sample variance (divide by $n-1$)
 - Biased variance estimator (divide by n)
 - b. Estimate:
 - Bias
 - Variance of each estimator
 - c. Plot bias versus n for both estimators.
 - d. Plot variance versus n .
4. Do a simulation to “verify” that sample variance is consistent for population variance.
5. Markov’s inequality states that for a nonnegative random variable X and any $a > 0$,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

In the following problems, verify this inequality empirically.

Let $X \sim \text{Exponential}(1)$.

- a. Estimate $\mathbb{E}[X]$ via simulation.
 - b. For $a = 0.5, 1, 2, 3, 5$:
 - Estimate $P(X \geq a)$.
 - Compute the Markov bound $\mathbb{E}[X]/a$.
 - c. Store results in a table with columns:
 - a
 - Estimated Probability
 - Markov Bound
 - d. Plot both quantities against a .
6. Let $X \sim \text{Uniform}(0, 2)$.
- a. Estimate $\mathbb{E}[X]$.
 - b. For $a = 0.5, 1, 1.5, 2, 3$:
 - Estimate $P(X \geq a)$.
 - Compute Markov bound.
 - c. Identify values of a for which the bound is trivial (greater than 1).

Worksheet 7

1. Verify that sample quantile is not parallelizable for the 0.25-quantile.

2. Verify that matrix multiplication is parallelizable, using the fact that for matrix $A \in \mathbb{R}^{n \times p}$ and matrix $B \in \mathbb{R}^{p \times m}$.

$$AB = \sum_{i=1}^n a_{.i} b_{i.},$$

Combined Questions

1. If matrix multiplication is done in a parallelized way given to c cores, what is the computational complexity for the work done in each core.
2. Verify Markov's inequality for a Gamma distribution, parallelizing replications.
3. Using draws from only $N(0, I)$ generate draws from $N(\mu, \Sigma)$ where
 - a. $\mu = 0$ and $\Sigma = LL^T$ where $L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 4 & 0 \\ 2 & 3 & 9 \end{pmatrix}$
 - b. $\mu = (10, 3, 5)^T$ and $\Sigma = \text{diag}(2, 5, 9)$.
 - c. $\mu = (0, 0, 0)^T$ and $\Sigma = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
4. Load the colored image of the dog in R and treat each pixel as a data point:
 - a. Run tSNE on the image with dimension = 1 and an appropriate perplexity
 - b. Convert the data back to an image and plot the lower dimensional image
 - c. Repeat the same for UMAP.
5. For hand-written digit question from the worksheet, using tSNE:
 - a. identify images that are possibly overlapping with other images in the embedding space
 - b. plot these images and understand why these are possibly lying in a different "clustering".
 - c. Hand draw your own number "eight" and upload the image in the same 16x16 pixel format. Include that as part of the handwritten dataset and assess whether its embedding matches the embeddings of the others "8"s.
6. We know that the sample variance with $1/(n-1)$ in the denominator is unbiased when the population is normal. For $F = N(0, 1)$, numerical check which of the $1/n$ vs $1/(n-1)$ sample variances have better MSE as a function of the sample size.

Worksheet 8 (Nelder-Mead)

1. Use the Nelder-Mead method to find the minimum of

$$f(x) = (x - 2)^2.$$

2. Use the Nelder-Mead method to find the minimum of

$$f(x, y) = x^2 + y^2$$

3. Use the Nelder-Mead method to find the minimum of

$$f(x, y) = x^2 + y^2 + 3 \sin(x) \sin(y)$$

4. Use the Nelder–Mead method to find the minimum of

$$f(x_1, x_2, x_3) = (x_1 - 1)^2 + (x_2 - 2)^2 + (x_3 - 3)^2$$

5. Use the Nelder–Mead method to find the minimum of

$$f(x, y) = (1 - x)^2 + 100(y - x^2)^2.$$

6. Consider the Bayesian lasso optimization problem:

$$\hat{\beta}_{\text{lasso}} = \arg \min_{\beta \in \mathbb{R}^d} \{ \|y - X\beta\|^2 + \lambda \|\beta\|_1 \}.$$

For the dataset below, use Nelder-Mead to obtain the lasso estimate for $\lambda = 10$.

- a. Choose the starting value carefully. What starting values can you use to ensure small number of iterations?

```
set.seed(123)
# Dimensions
n <- 50
p <- 3
# Design matrix
X <- matrix(rnorm(n * p), nrow = n, ncol = p)

# True coefficients
beta_true <- c(2.0, 0.0, -1.5)

# Generate response
y <- X %*% beta_true + rnorm(n, sd = sigma)
```

Worksheet 9

1. Load the dataset **quakes**, which contains 1,000 observations of earthquakes near Fiji. The variables are latitude (**lat**), longitude (**long**), depth (**depth**), magnitude (**mag**), and number of seismic stations (**stations**) reporting the event.
 - a. Determine the appropriate number of clusters k using the scree plot.
 - b. Apply the k-means clustering algorithm using the value of k obtained in part (a).
2. Load the dataset **Social_Network_Ads.csv** that contains information about users of a social network advertisement campaign. The response/label is the binary variable **Purchased**.
 - a. Split dataset into training and testing.
 - b. Implement k -NN algorithm for predictions and obtain prediction error.

3. Load the dataset **Salary_Data.csv** which contains Years Experience and Salary.
 - a. Fit $k = 2$ clusters. Based on the clustering, can you provide an interpretation for the clusters?
 - b. Fit $k = 3$ clusters. Provide an interpretation for this clustering.
 - c. Using scree plot choose the optimal number of clusters.
4. Load the dataset **CarPrice_Assignment.csv**, which contains various features of cars such as engine size, horsepower, mileage, dimensions, and the price of cars for different companies. It also includes a variable called symboling, treat symboling as the predict variable.
 - a. Split dataset into training and testing.
 - b. Implement k -NN on the full dimensions for predictions and compare with dimension reduced prediction error.

Worksheet 11 (Confidence Intervals)

1. $F = \text{Gamma}(\alpha, 5)$, $n = 50$ and estimator is sample mean for $\alpha = 2$.
 - a. Construct 95 % confidence interval for α .
 - b. Calculate the coverage probability.
 - c. For what values of n coverage probability is approximate 0.95.
2. $F_1 = N(\mu_1, 12)$, $n_1 = 50$ and $F_2 = N(\mu_2, 15)$, $n_2 = 60$. Sample means are estimator for μ_1 and μ_2 .
 - a. Construct 95 % confidence interval for $\mu_1 - \mu_2$.
 - b. Calculate the coverage probability.
3. A company wants to estimate the proportion of customers who prefer its new product over the old one. Suppose the true population proportion is $p = 0.6$. A random sample of size $n = 100$ is selected. Let $X \sim \text{Binomial}(n, p)$ be the number of customers who prefer the new product and the sample proportion is $\hat{p} = \frac{X}{n}$.
 - a. Construct 95 % confidence interval for p .
 - b. Calculate the coverage probability.
4. $F = \text{Exp}(\lambda)$, $n = 100$ and sample mean is the estimator for $\lambda = 4$.
 - a. Construct 95 % confidence interval for λ .
 - b. Calculate the coverage probability.
5. **Level α -test:** Just like confidence intervals, we can also verify p -value in Hypothesis testing. Consider the problem of testing $H_0 : \mu = 0$ versus $H_a : \mu \neq 0$ for a population $N(\mu, \sigma^2)$. Specifically, consider $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$.

The estimator for $\mu = \bar{X}_n$ and the estimator of σ^2 is the sample variance. The test statistic is

$$T_n = \frac{\bar{X}_n - 0}{\sqrt{\frac{\hat{\sigma}_n^2}{n}}}.$$

The null hypothesis is rejected if $|T_n| \geq t_{1-\alpha/2, n-1}$. Recall that α is chosen to ensure that

$$\Pr(\text{Reject } H_0 \mid H_0 \text{ is true}) = \alpha.$$

α is the type I error. Conduct a simulation a study to verify that the test maintains this type I error.

Worksheet 12 (LDA)

1. Generate two classes of data in $\mathbb{R}^2$ with the same covariance matrix. Fit an LDA model and compute the misclassification rate.
2. Generate a dataset with $d = 5$ features and two classes, assuming a common covariance matrix. Apply LDA and compute the misclassification rate.
3. Using the **iris dataset**: Split the data into training (70%) and testing (30%) sets - Fit an LDA model on the training data
Evaluate misclassification rate on the test data.
4. For the digits dataset, use LDA to classify between digits 0 and 7 (only).

Worksheet 13

1. Load the dataset **test_perform.csv** that contains information on a competitive exam collected from 1000 schools.
 - a. **studs**: number of students in the school that took the exam
 - b. **y**: average score from the number of students who took the exam
 - c. **teach**: an assessment score on the quality of teachers in that school
 - d. **budget**: an assessment score on the yearly budget of the school
 - e. **extra**: as assessment score on the amount of extra-curricular activites the school encourages.

Compute OLS estimator and compare it with ridge estimator.
2. Load the data **table.b1** in package **MPV** which contains data from the National Football League (the wrong football) from 1976.
 - a. Obtain the OLS estimator for the regression coefficients (with intercept).
 - b. Obtain the ridge estimator for the regression coefficients.
3. Consider the **cars** dataset in base R which contains information on the distance a car traveled after breaking. Obtain OLS estimator and compare it with ridge estimator.


```
library(MASS)
data(cars)
```

4. Simulate fake data according to the linear regression model $y = X\beta + \epsilon$. This can be done by the following,

```
set.seed(1)

n <- 100
p <- 6

beta.star <- c(2, 0.2, 4, 1, 3, .5)

# Design matrix
X <- matrix(rnorm(n*p), nrow = n, ncol = p)

# Generating response
y <- X %*% beta.star + rnorm(n, mean = 0, sd = 1)
```

- a. Obtain OLS estimator and Verify that OLS estimator is unbiased.
- b. Obtain ridge estimator for $\lambda = 4$ and find the biased of ridge estimator.